

ProjectSummary

Nicholas Cvercko

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Variability of Frequency/Positivity

Generate Tissue XY Values

Parameters varied for Simulations: The maxs list indicates the values for maximum X and Y values used for Kernel Window while the mins list indicates the minimum values used, and the kdis list is a dummy variable showing the distance between the maximum and minimum values (the range of x values used for a simulation were equal to the range of y values used in that same simulation so only the one range/distance is reported). 6 different min and max values were used with 20 simulations run for each.

```
maxs <- seq(9, 6.5, -0.5)
mins <- seq(1, 3.5, 0.5)
kdis <- maxs - mins
```

```
maxs
```

```
## [1] 9.0 8.5 8.0 7.5 7.0 6.5
```

```
mins
```

```
## [1] 1.0 1.5 2.0 2.5 3.0 3.5
```

```
kdis
```

```
## [1] 8 7 6 5 4 3
```

```
objs <- readRDS('GenTissXY_OBJS.rds')
pl <- readRDS('GenTissXY_pl.rds')
```

Below are the parameters that were used for the first simulation. The ymax, ymin, xmax, and xmin values for the tissue were changed for subsequent simulations while all other parameters were kept constant.

```
param <- ExtractParameters(objs[[1]][[1]], 'All')
```

```
## Holes has not yet been simulated
```

```
pv <- vector('list', length(param))
for (i in 1:length(param)){
  pv[[i]] <- unlist(param[[i]])
}
```

```
names(pv) <- names(param)
```

```
pv
```

```
## $Tissue
```

```
##      k  xmin  xmax  ymin  ymax sdmin sdmax
```

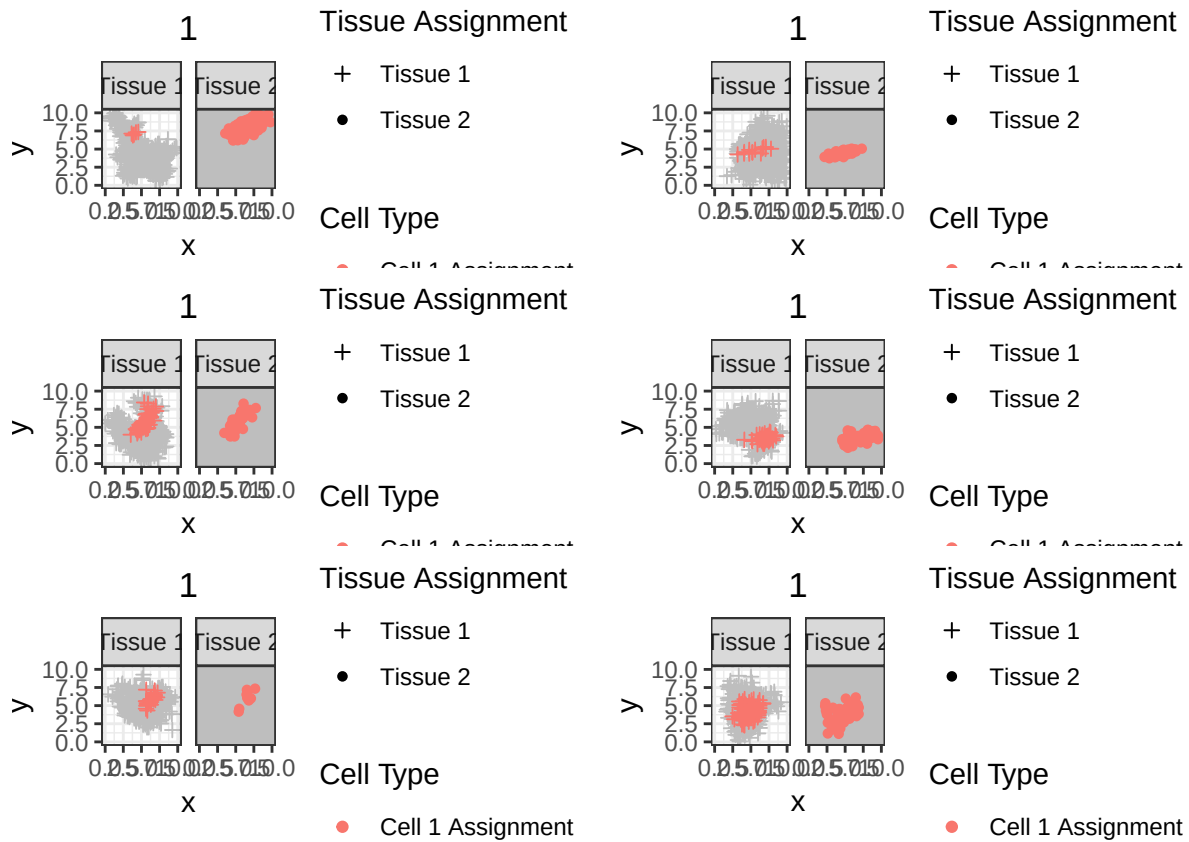
```
## 10.0  1.0   9.0   1.0   9.0   0.5   2.0
```

```
##
```

```
## $Holes
##      xmin      xmax      ymin      ymax      sadmin      sdmax hole_prob1
##      0.00     10.00      0.00     10.00      0.50      2.00      0.20
## hole_prob2
##      0.35
##
## $Cells
##      Cell Type 1.k      Cell Type 1.xmin      Cell Type 1.xmax      Cell Type 1.ymin
##      1.0              1.0              9.0              1.0
##      Cell Type 1.ymax      Cell Type 1.sadmin      Cell Type 1.sdmax      Cell Type 1.probs1
##      9.0              0.5              2.0              0.0
## Cell Type 1.probs2
##      1.0
```

Calculations of frequencies/positivities and boxplots.

```
set.seed(100)
pf <- (p1[[1]][[1]] + p1[[2]][[1]]) / (p1[[3]][[1]] + p1[[4]][[1]]) / (p1[[5]][[1]] + p1[[6]][[1]]); pf
```



```
ggsave("tissue_XY.png", plot = pf, dpi = 300, scale = 2)
```

```
ca <- vector('list', length(objs))
cpos <- vector('list', length(objs))
cfreq <- vector('list', length(objs))

for (i in 1:length(objs)) {
  ca[[i]] <- vector('list', length(objs[[i]]))
  cfreq[[i]] <- vector('list', length(objs[[i]]))
}
```

```

  cpos[[i]] <- vector('list', length(objs[[i]]))
}

for (i in 1:length(objs)) {
  for(j in 1:length(objs[[i]])) {
    ca[[i]][[j]] <- CreateSpatialList(objs[[i]][[j]][[1]][3:4])
    cfreq[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 1')/sum(ca[[i]][[j]][1] == 'Tissue 1')
    cpos[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 1')
  }
}

f1 <- vector('list', length(objs))
p1 <- vector('list', length(objs))
v.f1 <- vector('list', length(objs))
v.p1 <- vector('list', length(objs))
for (i in 1:length(f1)) {
  f1[[i]] <- unlist(cfreq[[i]])
  p1[[i]] <- unlist(cpos[[i]])
  v.f1[[i]] <- var(f1[[i]])
  v.p1[[i]] <- var(p1[[i]])
}

# create dataframes for pos and freq
d.f1 <- as.data.frame(f1)
d.p1 <- as.data.frame(p1)

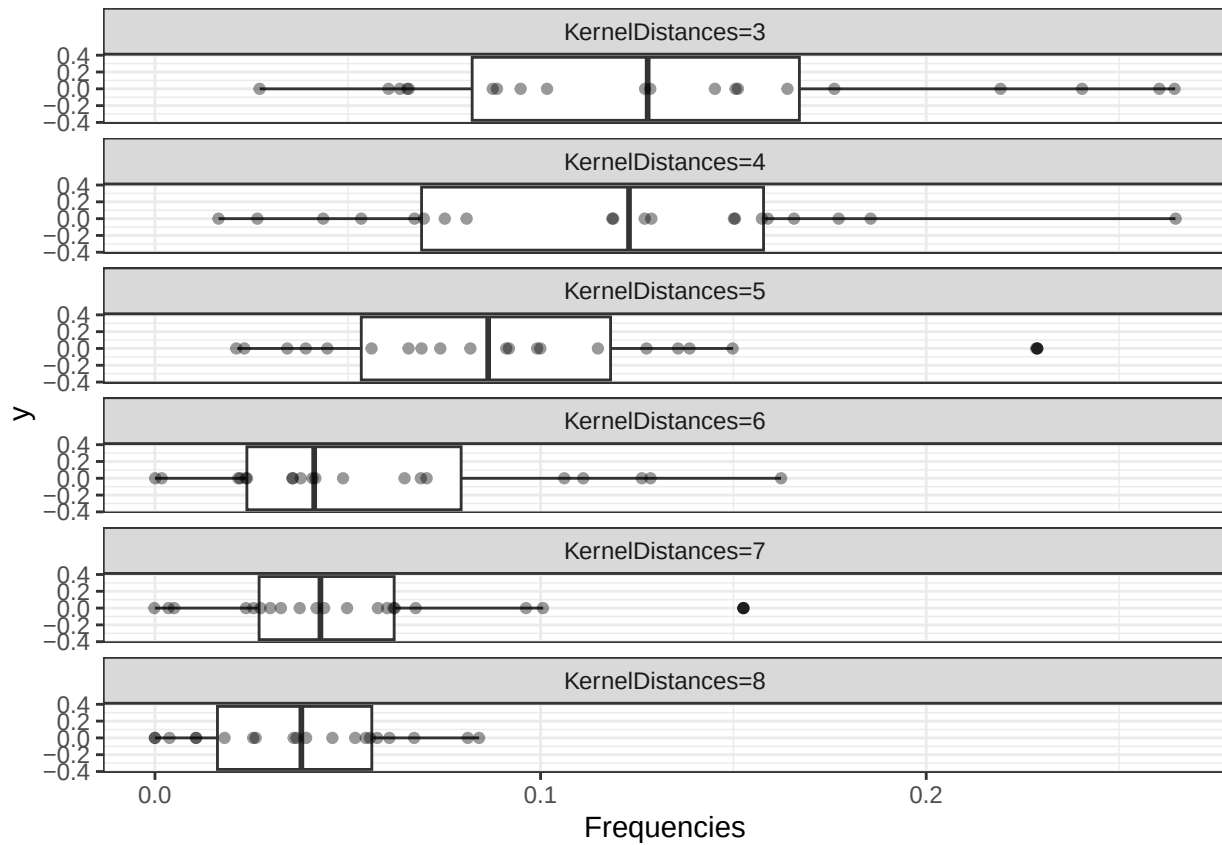
cn <- list(paste0('KernelDistances=', kdis))[[1]]
colnames(d.f1) <- cn
colnames(d.p1) <- cn

# convert to long frames
d.f1.long <- pivot_longer(d.f1,
  cols = all_of(cn),
  names_to = "kdis",
  values_to = "Frequencies")

d.p1.long <- pivot_longer(d.p1,
  cols = all_of(cn),
  names_to = "kdis",
  values_to = "Positives")

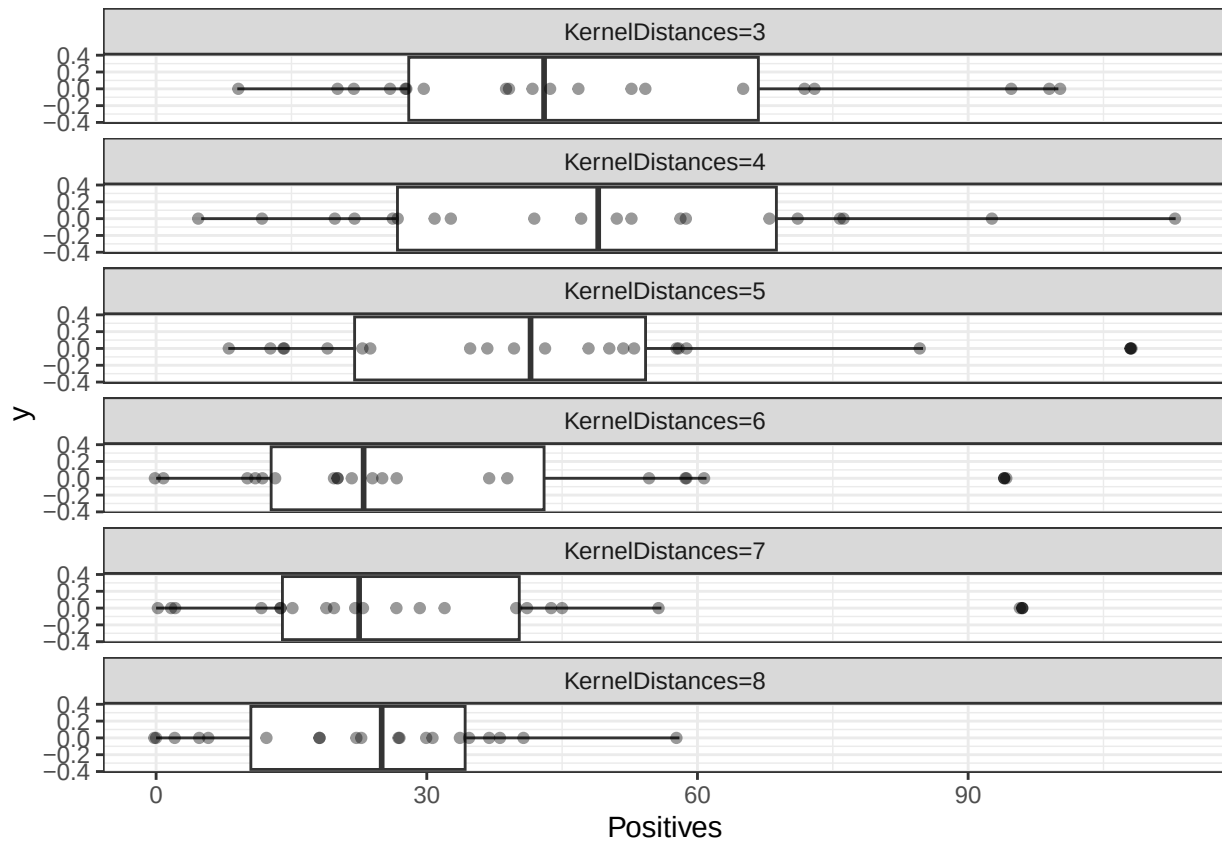
# create boxplots
freq.box <- ggplot(d.f1.long, aes(y= 0, x= Frequencies)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ kdis, ncol = 1) +
  theme_bw();freq.box

```



```
ggsave("GenTiss_xy_box_freq.png", plot = freq.box, dpi = 300, scale = 2)

pos.box <- ggplot(d.p1.long, aes(y= 0, x= Positives)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ kdis, ncol = 1) +
  theme_bw();pos.box
```



```
ggsave("GenTiss_xy_box_pos.png", plot = pos.box, dpi = 300, scale = 2)
```

Generate Tissue Kernel

Parameters varied for Simulations: The number of kernels used for the generate tissue command was changed. 6 different kernel values were used with 20 simulations run for each.

```
kr <- seq(1, 12, 2)

kr

## [1] 1 3 5 7 9 11

objs <- readRDS('tissue_kernel_OBJS.rds')
pl <- readRDS('tissue_kern_pl.rds')

ot <- vector('list', length(pl))
for (i in 1:length(pl)) {
  ot[[i]] <- pl[[i]][[1]]
}
```

Below are the parameters that were used for the first simulation. The value for the number of kernels for the tissue were changed for subsequent simulations while all other parameters were kept constant.

```
param <- ExtractParameters(objs[[1]][[1]], 'All')

## Holes has not yet been simulated

pv <- vector('list', length(param))
for (i in 1:length(param)){
```

```

pv[[i]] <- unlist(param[[i]])
}
names(pv) <- names(param)
pv

```

```

## $Tissue
##      k xmin  xmax ymin  ymax sdmin sdmax
##      1.0  2.5   7.5  2.5   7.5   0.5   2.0
##
## $Holes
##      xmin      xmax      ymin      ymax      sdmin      sdmax hole_prob1
##      0.00     10.00      0.00     10.00      0.50      2.00      0.20
## hole_prob2
##      0.35
##
## $Cells
##      Cell Type 1.k      Cell Type 1.xmin      Cell Type 1.xmax      Cell Type 1.ymin
##      1.0          2.5          7.5          2.5
##      Cell Type 1.ymax      Cell Type 1.sdmin      Cell Type 1.sdmax      Cell Type 1.probs1
##      7.5          0.5          2.0          0.0
## Cell Type 1.probs2
##      1.0

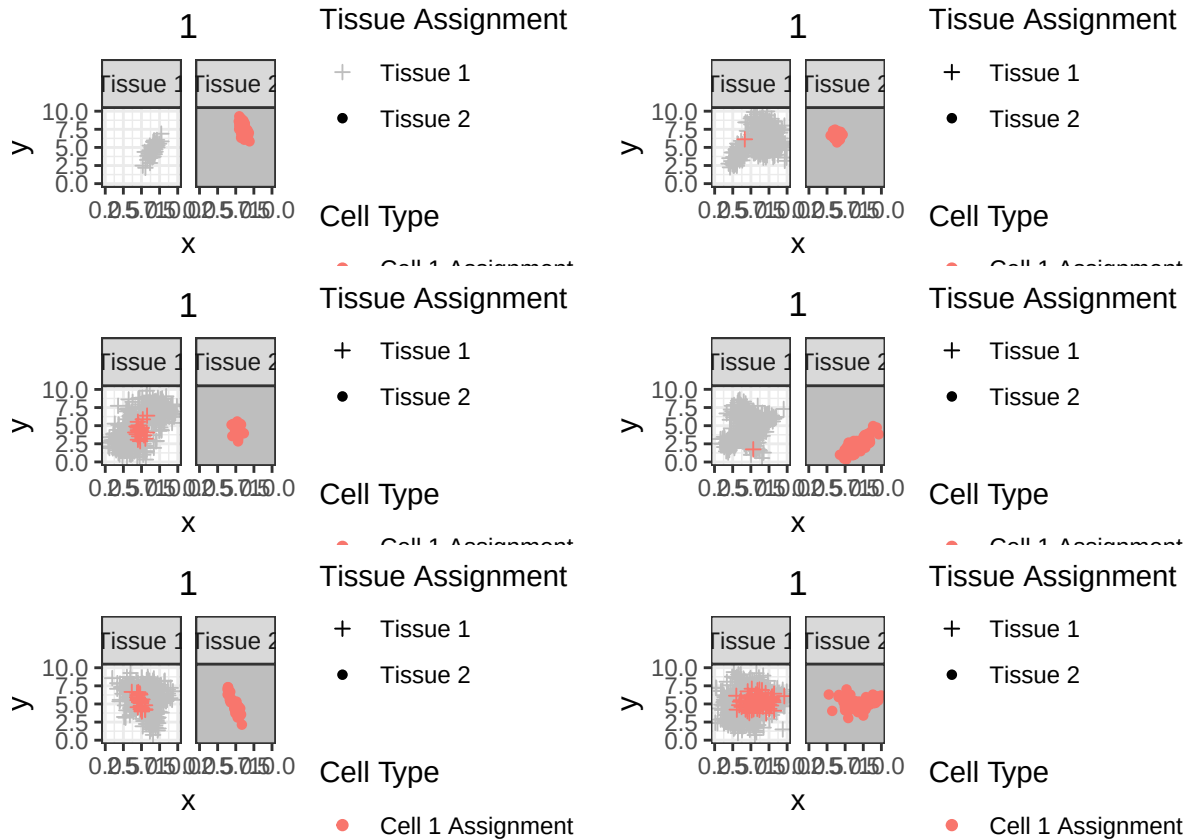
```

Calculations of frequencies/positivities and boxplots.

```

set.seed(100)
pf <- (ot[[1]] + ot[[2]]) / (ot[[3]] + ot[[4]]) / (ot[[5]] + ot[[6]]);pf

```



```

ggsave("tissue_kernel.png", plot = pf, dpi = 300, scale = 2)

## Saving 13 x 9 in image
ca <- vector('list', length(objs))
cpos <- vector('list', length(objs))
cfreq <- vector('list', length(objs))

for (i in 1:length(objs)) {
  ca[[i]] <- vector('list', length(objs[[i]]))
  cfreq[[i]] <- vector('list', length(objs[[i]]))
  cpos[[i]] <- vector('list', length(objs[[i]]))
}

for (i in 1:length(objs)) {
  for(j in 1:length(objs[[i]])) {
    ca[[i]][[j]] <- CreateSpatialList(objs[[i]][[j]])[[1]][3:4]
    cfreq[[i]][[j]] <- sum(ca[[i]][[j]][[2]] == 1 & ca[[i]][[j]][[1]] == 'Tissue 1')/sum(ca[[i]][[j]][[1]] == 'Tissue 1')
    cpos[[i]][[j]] <- sum(ca[[i]][[j]][[2]] == 1 & ca[[i]][[j]][[1]] == 'Tissue 1')
  }
}

f1 <- vector('list', length(objs))
p1 <- vector('list', length(objs))
v.f1 <- vector('list', length(objs))
v.p1 <- vector('list', length(objs))
for (i in 1:length(f1)) {
  f1[[i]] <- unlist(cfreq[[i]])
  p1[[i]] <- unlist(cpos[[i]])
  v.f1[[i]] <- var(f1[[i]])
  v.p1[[i]] <- var(p1[[i]])
}

# create dataframes for pos and freq
d.f1 <- as.data.frame(f1)
d.p1 <- as.data.frame(p1)

cn <- list(paste0('K=0', kr[1:(length(kr)-1)]))[[1]]
cn <- c(cn, 'K=11')
colnames(d.f1) <- cn
colnames(d.p1) <- cn

# convert to long frames

d.f1.long <- pivot_longer(d.f1,
                          cols = all_of(cn),
                          names_to = "kern",
                          values_to = "Frequencies")

d.p1.long <- pivot_longer(d.p1,
                          cols = all_of(cn),
                          names_to = "kern",

```

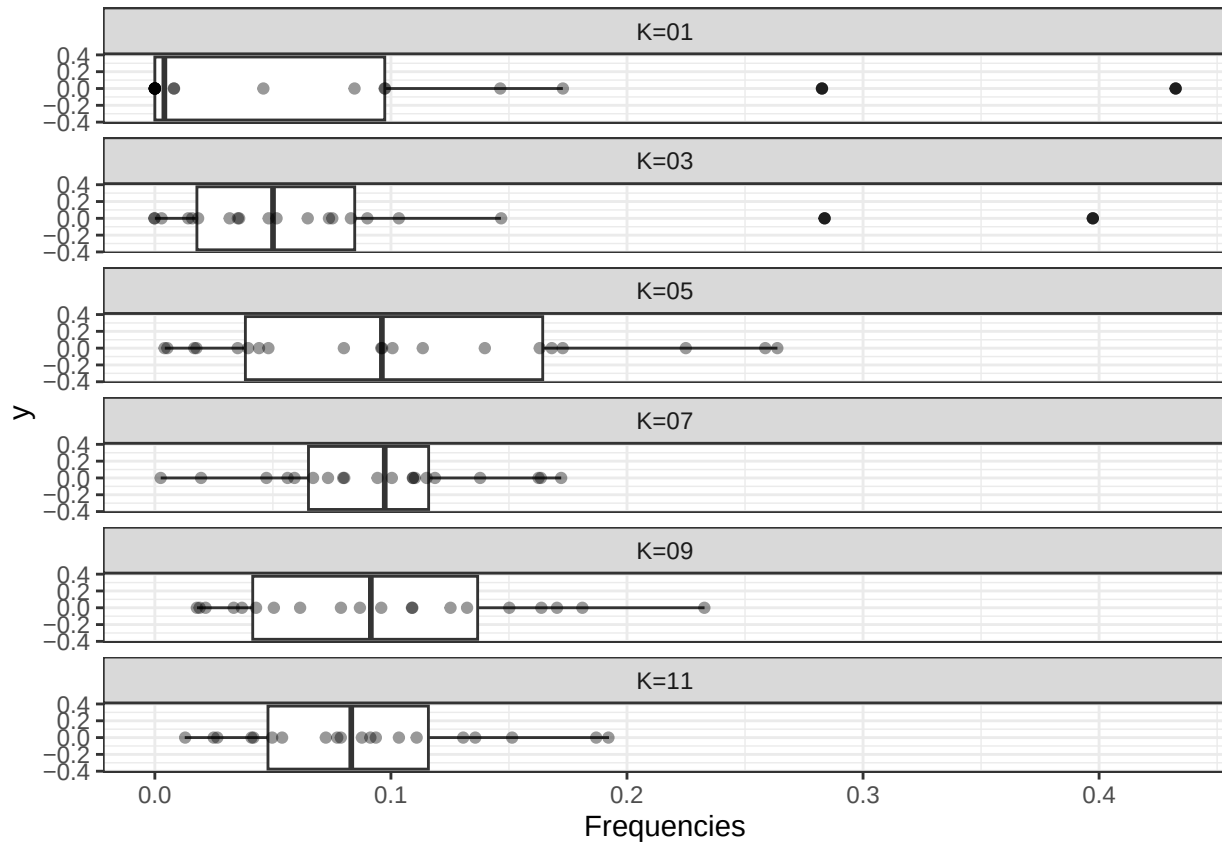
```

values_to = "Positives")

# create boxplots

freq.box <- ggplot(d.f1.long, aes(y= 0, x= Frequencies)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ kern, ncol = 1) +
  theme_bw();freq.box

```



```

ggsave("gentiss_kern_box_freq.png", plot = freq.box, dpi = 300, scale = 2)

```

```

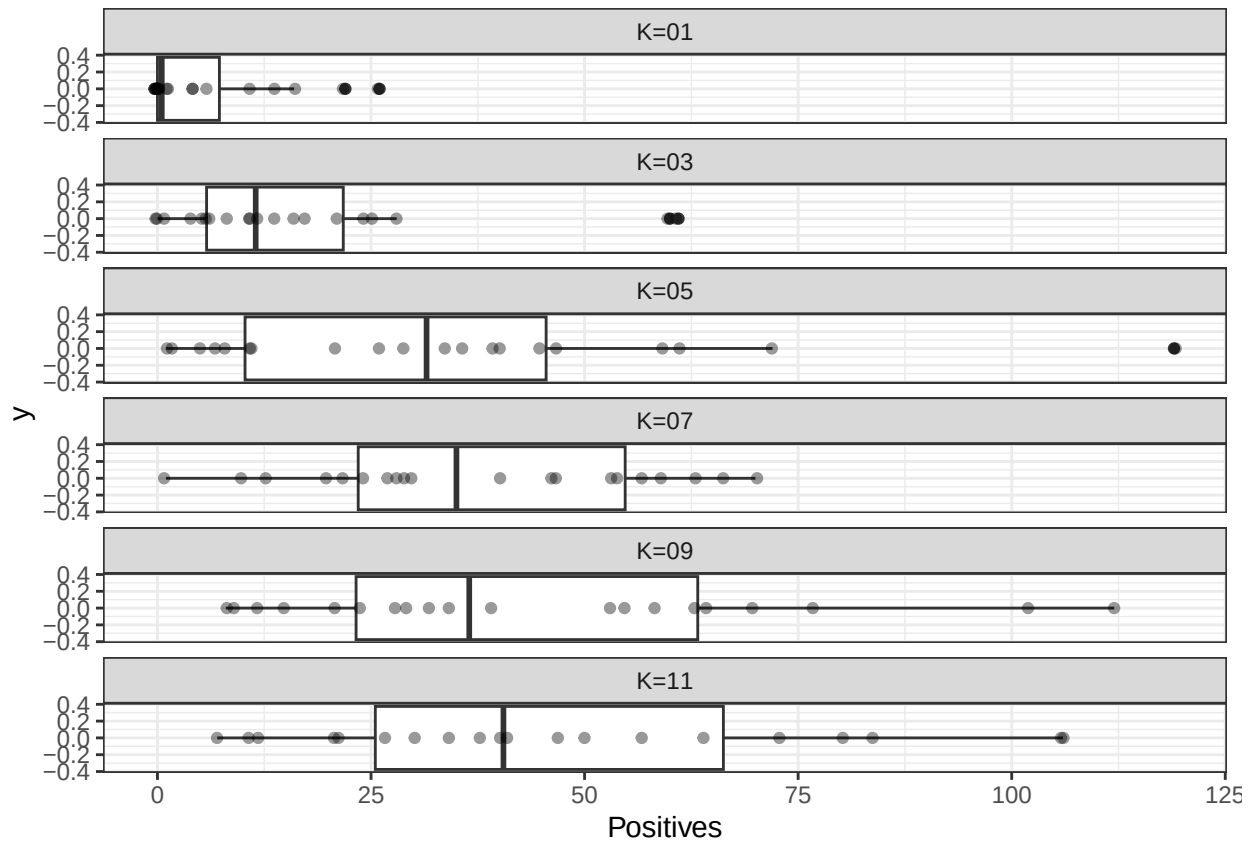
## Saving 13 x 9 in image

```

```

pos.box <- ggplot(d.p1.long, aes(y= 0, x= Positives)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ kern, ncol = 1) +
  theme_bw();pos.box

```



```
ggsave("gentiss_kern_box_pos.png", plot = pos.box, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

Generate Cell Positivity Probabilities

Parameters varied for Simulations: The minimum probability for the generation of positive cell assignments to cell type 1 were varied. 6 different probabilities were used with 20 simulations run for each.

```
probs <- seq(1, 0, -0.2)
probs
```

```
## [1] 1.0 0.8 0.6 0.4 0.2 0.0
```

```
objs <- readRDS('GenPosProbs_OBJS.rds')
pl <- readRDS('GenPosProbs_pl.rds')
```

Below are the parameters that were used for the first simulation. The value for the minimum probability of being assigned to cell type 1 was changed for subsequent simulations while all other parameters were kept constant.

```
param <- ExtractParameters(objs[[1]][[1]], 'All')
```

```
## Holes has not yet been simulated
```

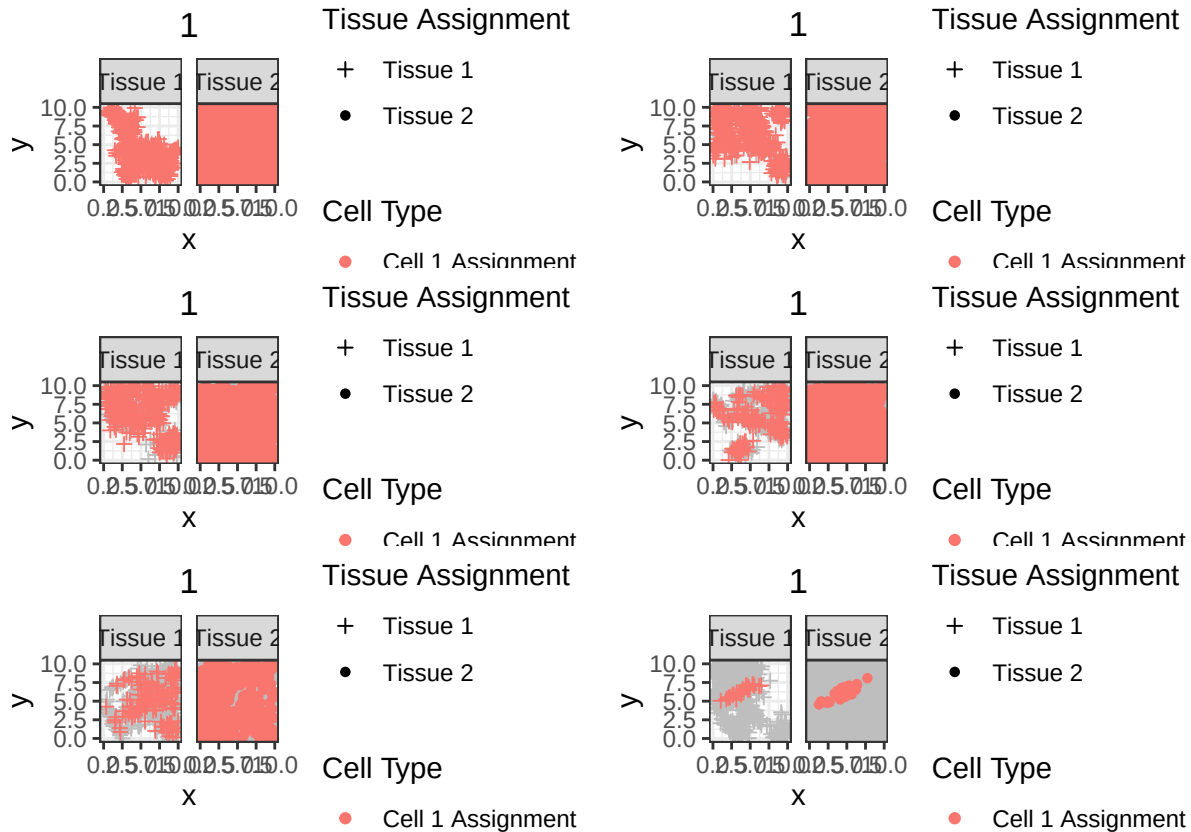
```
pv <- vector('list', length(param))
for (i in 1:length(param)){
  pv[[i]] <- unlist(param[[i]])
}
names(pv) <- names(param)
```

```
pv
```

```
## $Tissue
##      k  xmin  xmax  ymin  ymax  sdmin  sdmax
## 10.0   0.0  10.0   0.0  10.0   0.5   2.0
##
## $Holes
##      xmin      xmax      ymin      ymax      sdmin      sdmax  hole_prob1
##      0.00      10.00      0.00      10.00      0.50      2.00      0.20
## hole_prob2
##      0.35
##
## $Cells
##      Cell Type 1.k  Cell Type 1.xmin  Cell Type 1.xmax  Cell Type 1.ymin
##                  1.0                2.5                7.5                2.5
##      Cell Type 1.ymax  Cell Type 1.sdmin  Cell Type 1.sdmax  Cell Type 1.probs1
##                  7.5                0.5                2.0                1.0
## Cell Type 1.probs2
##                  1.0
```

Calculations of frequencies/positivities and boxplots.

```
pf <- (p1[[1]][[1]] + p1[[2]][[1]]) / (p1[[3]][[1]] + p1[[4]][[1]]) / (p1[[5]][[1]] + p1[[6]][[1]]); pf
```



```
ggsave("GenProbs.png", plot = pf, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

```

ca <- vector('list', length(objs))
cpos <- vector('list', length(objs))
cfreq <- vector('list', length(objs))

for (i in 1:length(objs)) {
  ca[[i]] <- vector('list', length(objs[[i]]))
  cfreq[[i]] <- vector('list', length(objs[[i]]))
  cpos[[i]] <- vector('list', length(objs[[i]]))
}

for (i in 1:length(objs)) {
  for(j in 1:length(objs[[i]])) {
    ca[[i]][[j]] <- CreateSpatialList(objs[[i]][[j]])[[1]][3:4]
    cfreq[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2')/sum(ca[[i]][[j]][1] == 1)
    cpos[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2')
  }
}

f1 <- vector('list', length(objs))
p1 <- vector('list', length(objs))
v.f1 <- vector('list', length(objs))
v.p1 <- vector('list', length(objs))
for (i in 1:length(f1)) {
  f1[[i]] <- unlist(cfreq[[i]])
  p1[[i]] <- unlist(cpos[[i]])
  v.f1[[i]] <- var(f1[[i]])
  v.p1[[i]] <- var(p1[[i]])
}

# create dataframes for pos and freq
d.f1 <- as.data.frame(f1)
d.p1 <- as.data.frame(p1)

cn <- list(paste0('MinProb=', probs))[[1]]
colnames(d.f1) <- cn
colnames(d.p1) <- cn

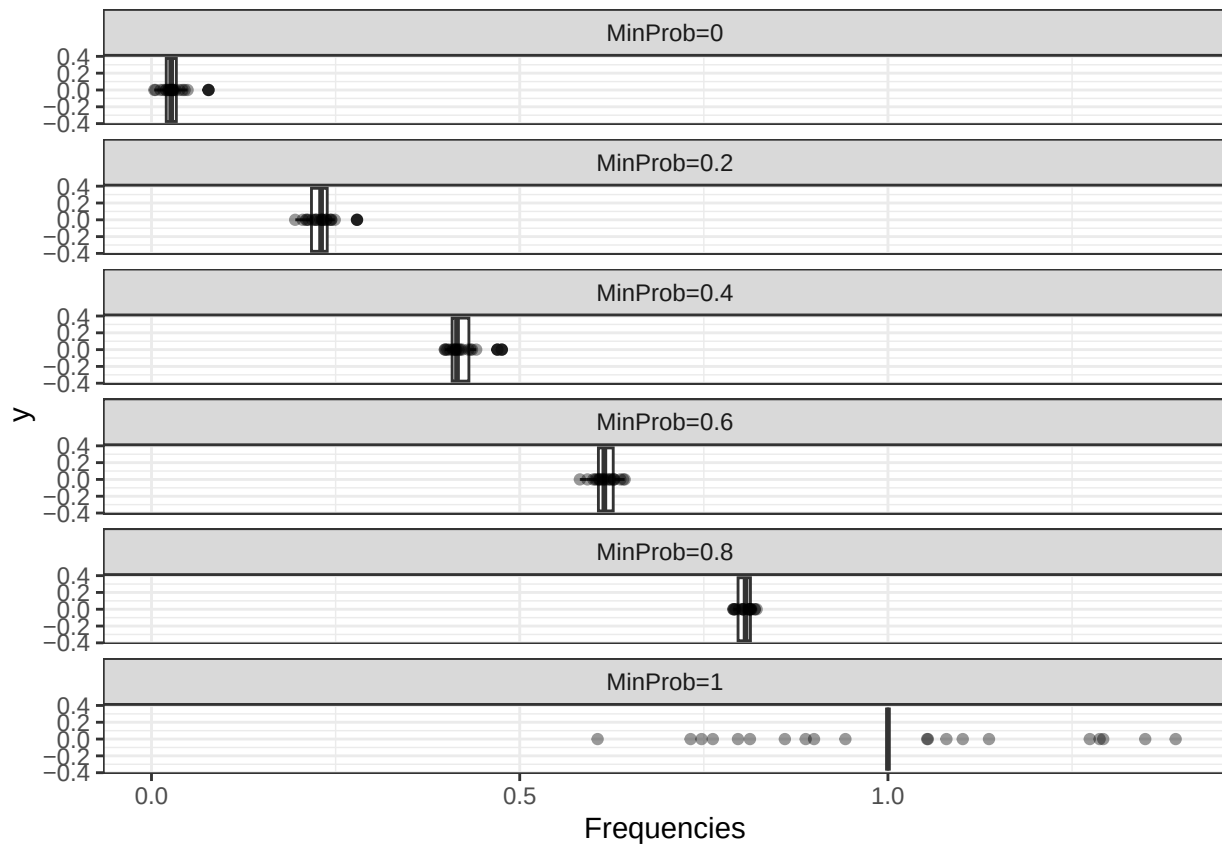
# convert to long frames
d.f1.long <- pivot_longer(d.f1,
                          cols = all_of(cn),
                          names_to = "p",
                          values_to = "Frequencies")

d.p1.long <- pivot_longer(d.p1,
                          cols = all_of(cn),
                          names_to = "p",
                          values_to = "Positives")

# create boxplots

```

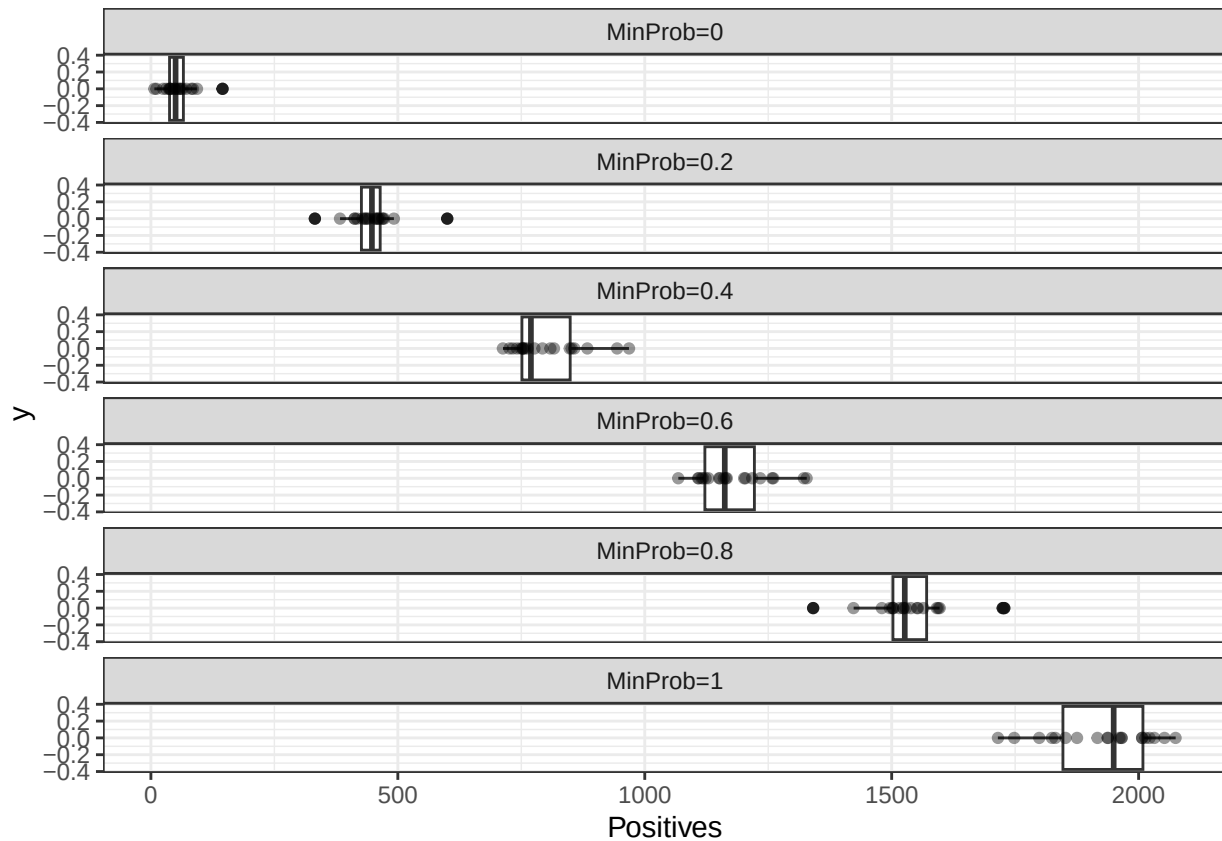
```
freq.box <- ggplot(d.f1.long, aes(y= 0, x= Frequencies)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ p, ncol = 1) +
  theme_bw();freq.box
```



```
ggsave("gpos_probs_box_freq.png", plot = freq.box, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

```
pos.box <- ggplot(d.p1.long, aes(y= 0, x= Positives)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ p, ncol = 1) +
  theme_bw();pos.box
```



```
ggsave("gpos_probs_box_pos.png", plot = pos.box, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

Generate Cell Positivity Kernel

Parameters varied for Simulations: The kernel number for the generation of positive cell assignments to cell type 1 were varied. 6 different kernel values were used with 20 simulations run for each.

```
kern <- seq(1, 16, 3);kern
```

```
## [1] 1 4 7 10 13 16
```

```
objs <- readRDS('tissue_kernel_OBJS.rds')
pl <- readRDS('tissue_kern_pl.rds')
```

Below are the parameters that were used for the first simulation. The value for the number of kernels for the generation of cell positivity to cell type 1 was changed for subsequent simulations while all other parameteres were kept constant.

```
param <- ExtractParameters(objs[[1]][[1]], 'All')
```

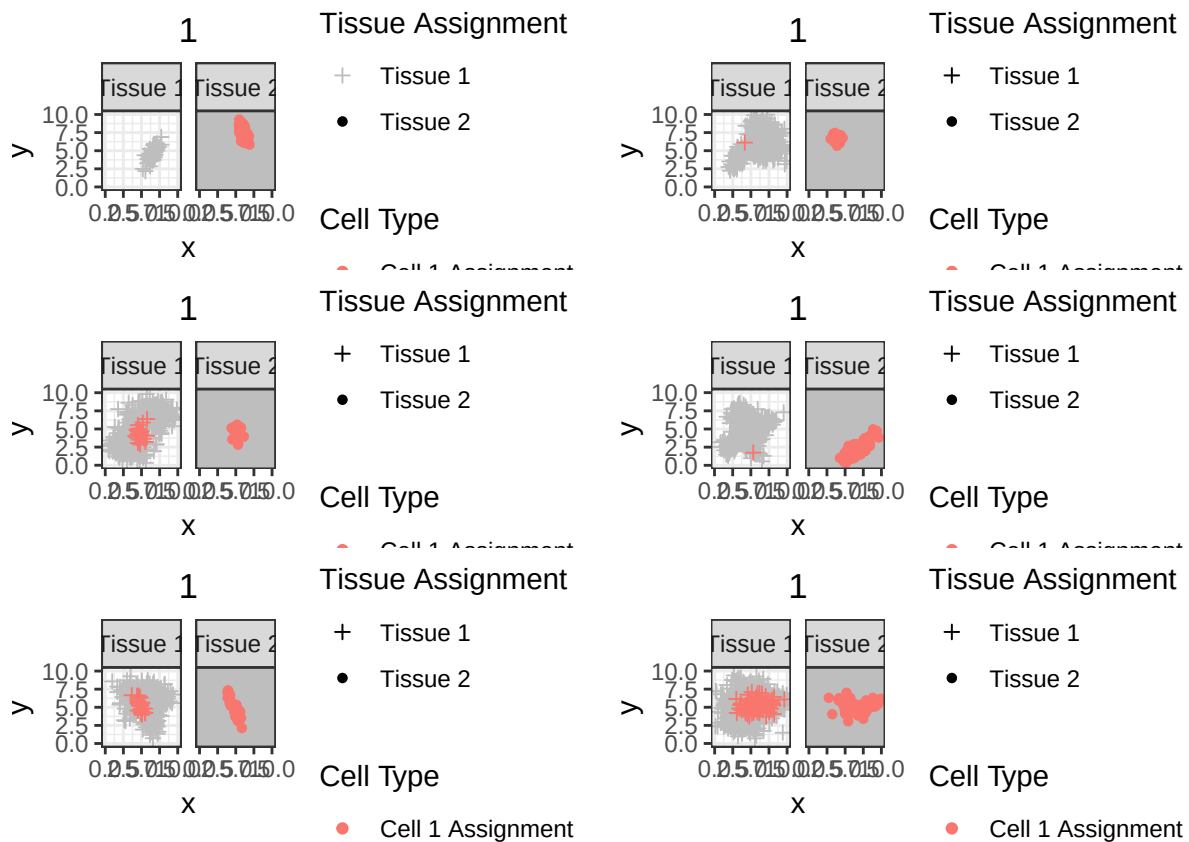
```
## Holes has not yet been simulated
```

```
pv <- vector('list', length(param))
for (i in 1:length(param)){
  pv[[i]] <- unlist(param[[i]])
}
names(pv) <- names(param)
pv
```

```
## $Tissue
##      k xmin  xmax ymin  ymax sdmin sdmax
##    1.0  2.5   7.5   2.5   7.5   0.5   2.0
##
## $Holes
##      xmin      xmax      ymin      ymax      sdmin      sdmax hole_prob1
##      0.00      10.00      0.00      10.00      0.50      2.00      0.20
## hole_prob2
##      0.35
##
## $Cells
##      Cell Type 1.k      Cell Type 1.xmin      Cell Type 1.xmax      Cell Type 1.ymin
##              1.0              2.5              7.5              2.5
##      Cell Type 1.ymax      Cell Type 1.sdmin      Cell Type 1.sdmax      Cell Type 1.probs1
##              7.5              0.5              2.0              0.0
## Cell Type 1.probs2
##              1.0
```

Calculations of frequencies/positivities and boxplots.

```
pf <- (p1[[1]][[1]] + p1[[2]][[1]]) / (p1[[3]][[1]] + p1[[4]][[1]]) / (p1[[5]][[1]] + p1[[6]][[1]]); pf
```



```
ggsave("GenKern.png", plot = pf, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

```
ca <- vector('list', length(objs))
cpos <- vector('list', length(objs))
cfreq <- vector('list', length(objs))
```

```

for (i in 1:length(objs)) {
  ca[[i]] <- vector('list', length(objs[[i]]))
  cfreq[[i]] <- vector('list', length(objs[[i]]))
  cpos[[i]] <- vector('list', length(objs[[i]]))
}

for (i in 1:length(objs)) {
  for(j in 1:length(objs[[i]])) {
    ca[[i]][[j]] <- CreateSpatialList(objs[[i]][[j]])[[1]][3:4]
    cfreq[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2')/sum(ca[[i]][[j]][1] == 'Tissue 2')
    cpos[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2')
  }
}

f1 <- vector('list', length(objs))
p1 <- vector('list', length(objs))
v.f1 <- vector('list', length(objs))
v.p1 <- vector('list', length(objs))
for (i in 1:length(f1)) {
  f1[[i]] <- unlist(cfreq[[i]])
  p1[[i]] <- unlist(cpos[[i]])
  v.f1[[i]] <- var(f1[[i]])
  v.p1[[i]] <- var(p1[[i]])
}

# create dataframes for pos and freq
d.f1 <- as.data.frame(f1)
d.p1 <- as.data.frame(p1)

cn <- list(paste0('K=0', kern[1:(length(kern)-3)]))[[1]]
cn <- c(cn, list(paste0('K=', kern[4:length(kern)]))[[1]])
colnames(d.f1) <- cn
colnames(d.p1) <- cn

# convert to long frames

d.f1.long <- pivot_longer(d.f1,
  cols = all_of(cn),
  names_to = "kern",
  values_to = "Frequencies")

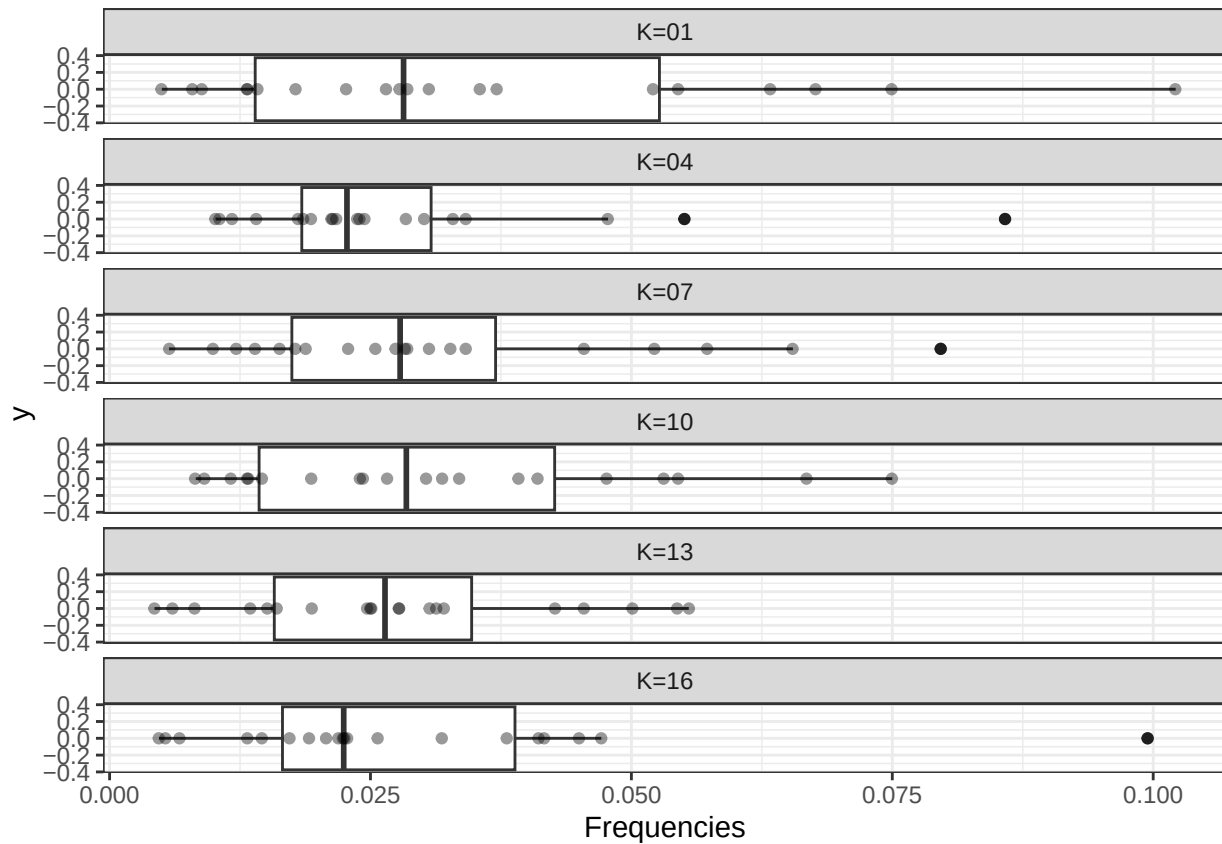
d.p1.long <- pivot_longer(d.p1,
  cols = all_of(cn),
  names_to = "kern",
  values_to = "Positives")

# create boxplots

freq.box <- ggplot(d.f1.long, aes(y= 0, x= Frequencies)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +

```

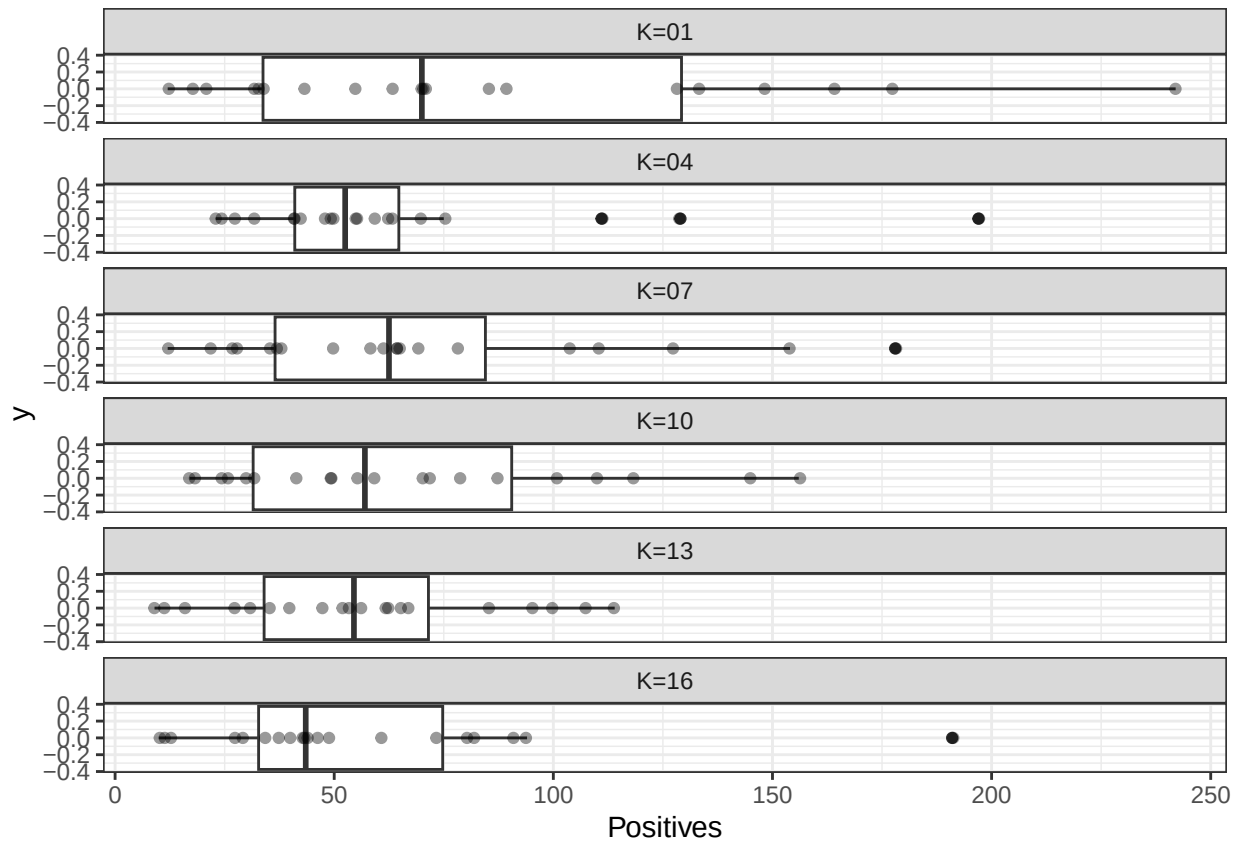
```
facet_wrap(~ kern, ncol = 1) +
theme_bw();freq.box
```



```
ggsave("gpos_kern_box_freq.png", plot = freq.box, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

```
pos.box <- ggplot(d.p1.long, aes(y= 0, x= Positives)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ kern, ncol = 1) +
  theme_bw();pos.box
```



```
ggsave("gpos_kern_box_pos.png", plot = pos.box, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

Generate Cell Positivity Maximum Kernel Standard Deviation

Parameters varied for Simulations: The maximum standard deviation values for the generation of positive cell assignments to cell type 1 were varied. 6 different standard deviation values were used with 20 simulations run for each.

```
maxs <- c(3, 2, 1.75, 1.5, 1, 0.75)
```

```
objs <- readRDS('sdmaxCentered20BJS.rds')
pl <- readRDS('sdmaxCentered2_pl.rds')
```

Below are the parameters that were used for the first simulation. The value for the maximum standard deviation for generation cell positivity to cell type 1 was changed for subsequent simulations while all other parameters were kept constant.

```
param <- ExtractParameters(objs[[1]][[1]], 'All')
```

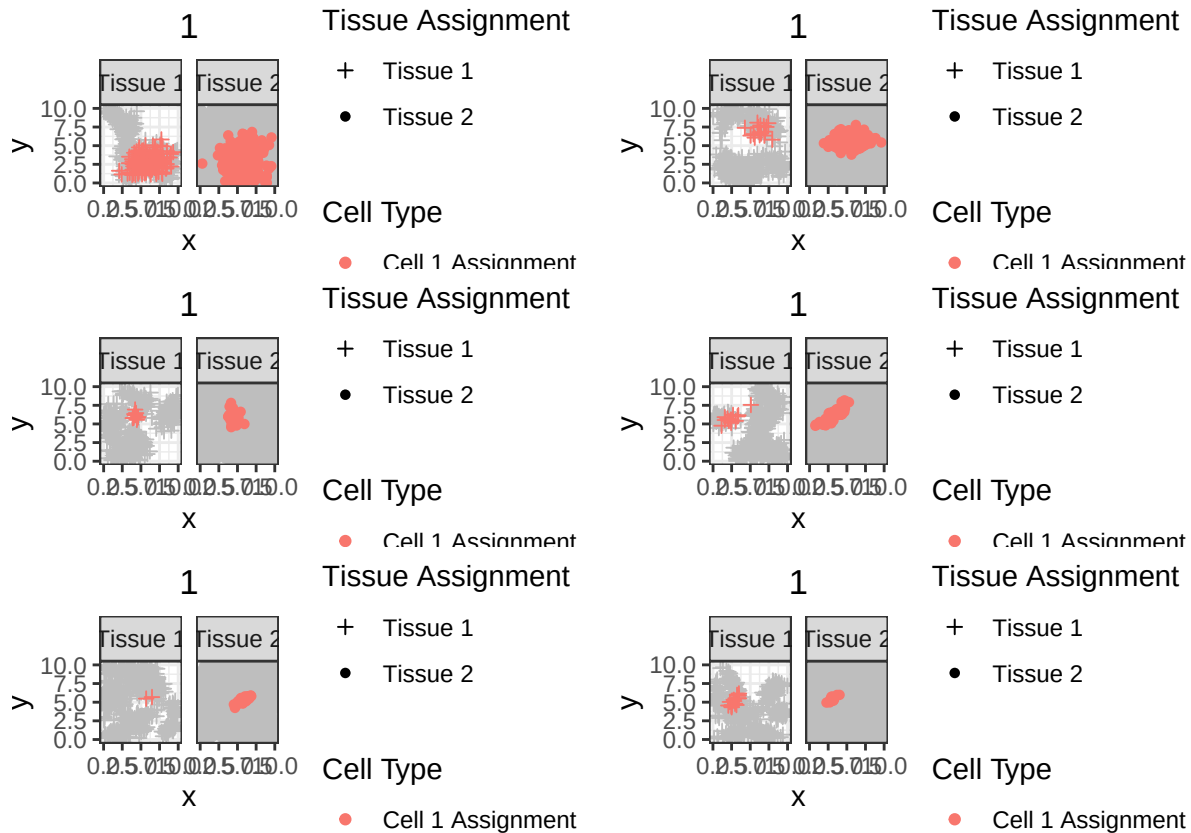
```
## Holes has not yet been simulated
```

```
pv <- vector('list', length(param))
for (i in 1:length(param)){
  pv[[i]] <- unlist(param[[i]])
}
names(pv) <- names(param)
pv
```

```
## $Tissue
##      k  xmin  xmax  ymin  ymax  sdmin  sdmax
## 10.0   0.0  10.0   0.0  10.0   0.5   2.0
##
## $Holes
##      xmin      xmax      ymin      ymax      sdmin      sdmax hole_prob1
##      0.00      10.00      0.00      10.00      0.50      2.00      0.20
## hole_prob2
##      0.35
##
## $Cells
##      Cell Type 1.k  Cell Type 1.xmin  Cell Type 1.xmax  Cell Type 1.ymin
##              1.0          2.5          7.5          2.5
##      Cell Type 1.ymax  Cell Type 1.sdmin  Cell Type 1.sdmax  Cell Type 1.probs1
##              7.5          0.5          3.0          0.0
## Cell Type 1.probs2
##              1.0
```

Calculations of frequencies/positivities and boxplots.

```
pf <- (p1[[1]][[1]] + p1[[2]][[1]]) / (p1[[3]][[1]] + p1[[4]][[1]]) / (p1[[5]][[1]] + p1[[6]][[1]]); pf
```



```
ggsave("GenPosMaxSD.png", plot = pf, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

```
ca <- vector('list', length(objs))
cpos <- vector('list', length(objs))
cfreq <- vector('list', length(objs))
```

```

for (i in 1:length(objs)) {
  ca[[i]] <- vector('list', length(objs[[i]]))
  cfreq[[i]] <- vector('list', length(objs[[i]]))
  cpos[[i]] <- vector('list', length(objs[[i]]))
}

for (i in 1:length(objs)) {
  for(j in 1:length(objs[[i]])) {
    ca[[i]][[j]] <- CreateSpatialList(objs[[i]][[j]])[[1]][3:4]
    cfreq[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2')/sum(ca[[i]][[j]][1] == 'Tissue 2')
    cpos[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2')
  }
}

f1 <- vector('list', length(objs))
p1 <- vector('list', length(objs))
v.f1 <- vector('list', length(objs))
v.p1 <- vector('list', length(objs))
for (i in 1:length(f1)) {
  f1[[i]] <- unlist(cfreq[[i]])
  p1[[i]] <- unlist(cpos[[i]])
  v.f1[[i]] <- var(f1[[i]])
  v.p1[[i]] <- var(p1[[i]])
}

# create dataframes for pos and freq
d.f1 <- as.data.frame(f1)
d.p1 <- as.data.frame(p1)

cn <- list(paste0('sdmax', maxs))[[1]]
colnames(d.f1) <- cn
colnames(d.p1) <- cn

# convert to long frames

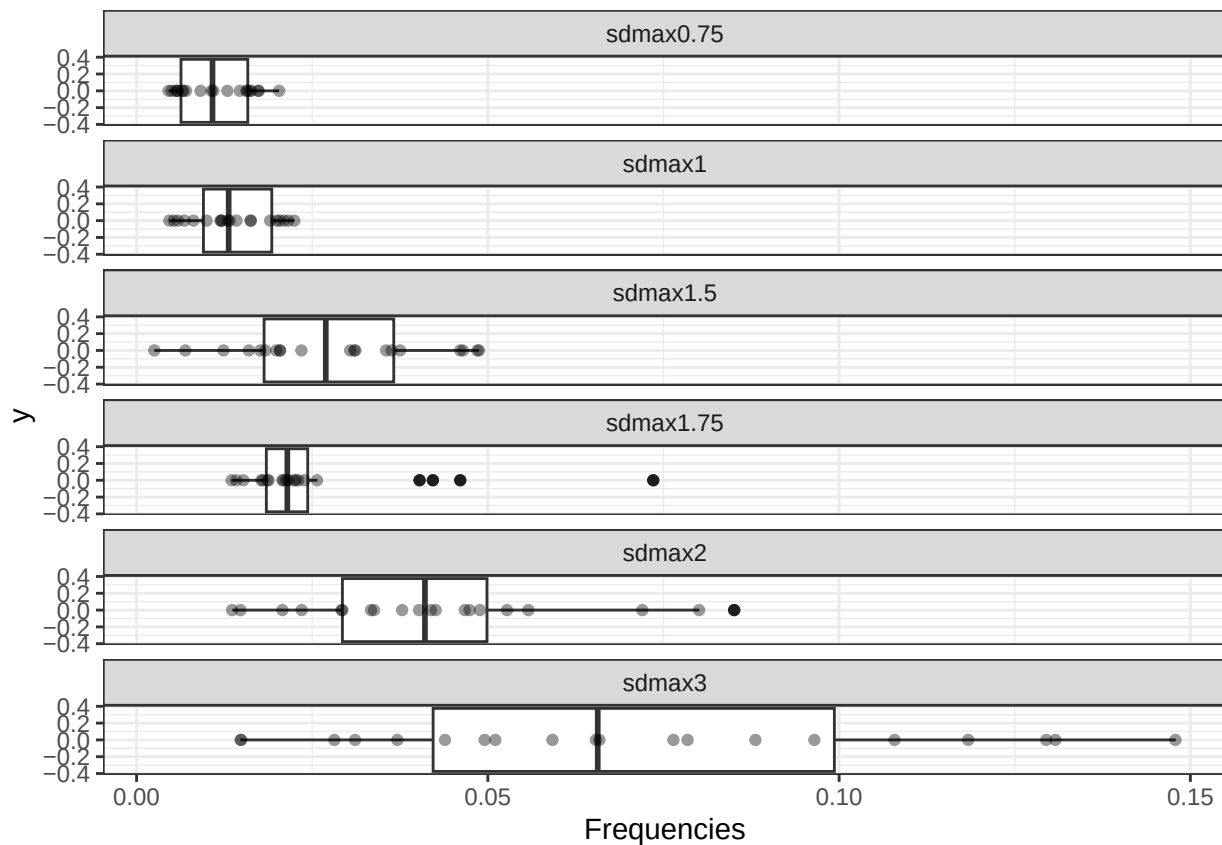
d.f1.long <- pivot_longer(d.f1,
  cols = all_of(cn),
  names_to = "sdmax",
  values_to = "Frequencies")

d.p1.long <- pivot_longer(d.p1,
  cols = all_of(cn),
  names_to = "sdmax",
  values_to = "Positives")

# create boxplots

freq.box <- ggplot(d.f1.long, aes(y= 0, x= Frequencies)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ sdmax, ncol = 1) +
  theme_bw();freq.box

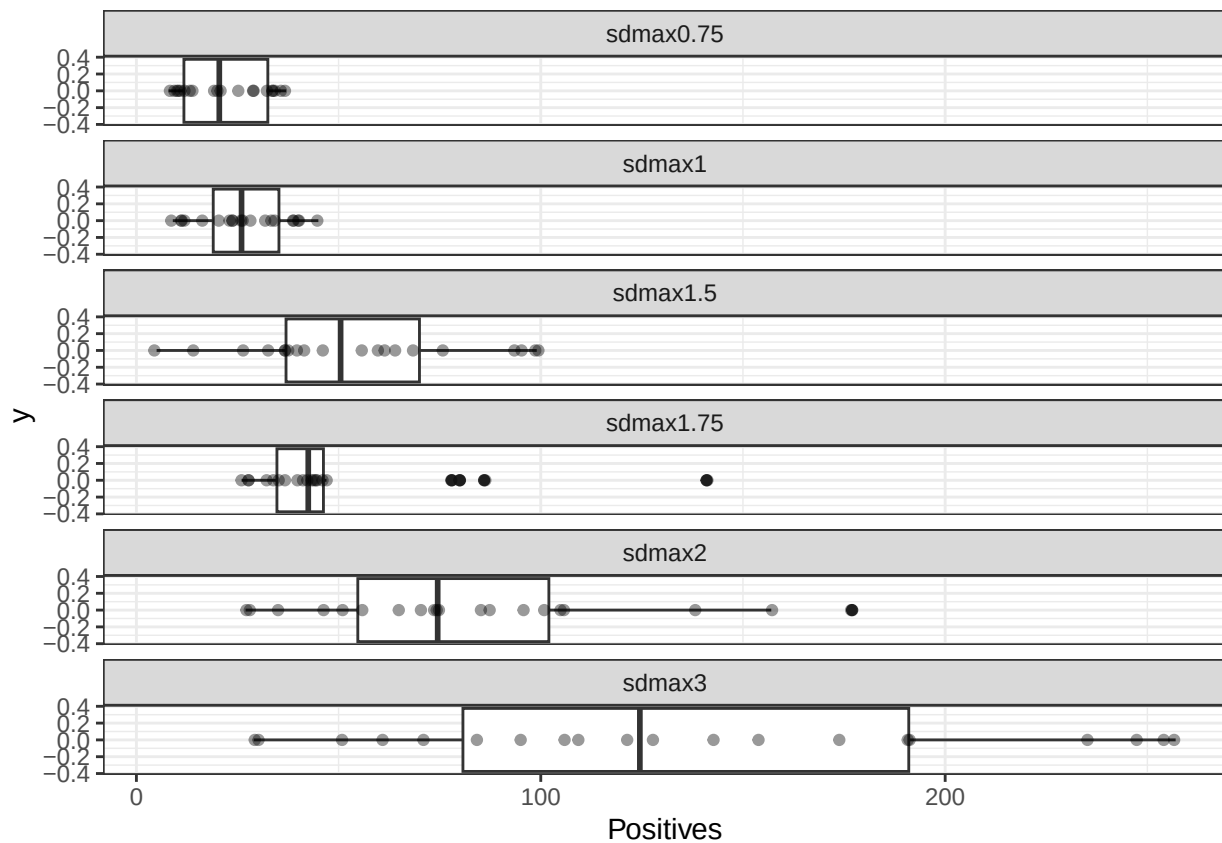
```



```
ggsave("gpos_sdmax_box_freq.png", plot = freq.box, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

```
pos.box <- ggplot(d.p1.long, aes(y= 0, x= Positives)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ sdmax, ncol = 1) +
  theme_bw();pos.box
```



```
ggsave("gpos_sdmax_box_pos.png", plot = pos.box, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

Simulation Seeds

Control

At first the same seed was used for all simulations to determine if there was baseline variability in the package. 20 simulations were run for 6 groups of simulations.

```
set.seed(800)
```

```
objs <- readRDS('StartControl10BJS.rds')
```

Below are the parameters that were used for the first simulation. All of these parameters were kept constant.

```
param <- ExtractParameters(objs[[1]][[1]], 'All')
```

```
## Holes has not yet been simulated
```

```
pv <- vector('list', length(param))
for (i in 1:length(param)){
  pv[[i]] <- unlist(param[[i]])
}
names(pv) <- names(param)
pv
```

```
## $Tissue
```

```
##      k xmin xmax ymin ymax sadmin sdmax
```

```
## 10.0 0.0 10.0 0.0 10.0 0.5 2.0
##
## $Holes
##      xmin      xmax      ymin      ymax      sdmin      sdmax hole_prob1
##      0.00      10.00      0.00      10.00      0.50      2.00      0.20
## hole_prob2
##      0.35
##
## $Cells
##      Cell Type 1.k      Cell Type 1.xmin      Cell Type 1.xmax      Cell Type 1.ymin
##              1.0              0.0              10.0              0.0
##      Cell Type 1.ymax      Cell Type 1.sdmin      Cell Type 1.sdmax      Cell Type 1.probs1
##              10.0              0.5              2.0              0.0
## Cell Type 1.probs2
##              1.0
```

Calculations of frequencies/positivities and boxplots.

```
ca <- vector('list', length(objs))
cpos <- vector('list', length(objs))
cfreq <- vector('list', length(objs))

for (i in 1:length(objs)) {
  ca[[i]] <- vector('list', length(objs[[i]]))
  cfreq[[i]] <- vector('list', length(objs[[i]]))
  cpos[[i]] <- vector('list', length(objs[[i]]))
}

for (i in 1:length(objs)) {
  for(j in 1:length(objs[[i]])) {
    ca[[i]][[j]] <- CreateSpatialList(objs[[i]][[j]])[[1]][3:4]
    cfreq[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2')/sum(ca[[i]][[j]][1] == 'Tissue 2')
    cpos[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2')
  }
}

f1 <- vector('list', length(objs))
p1 <- vector('list', length(objs))
v.f1 <- vector('list', length(objs))
v.p1 <- vector('list', length(objs))
for (i in 1:length(f1)) {
  f1[[i]] <- unlist(cfreq[[i]])
  p1[[i]] <- unlist(cpos[[i]])
  v.f1[[i]] <- var(f1[[i]])
  v.p1[[i]] <- var(p1[[i]])
}

# create dataframes for pos and freq
d.f1 <- as.data.frame(f1)
d.p1 <- as.data.frame(p1)
```

```

cn <- list(paste0('sim_set', 1:6))[[1]]
colnames(d.f1) <- cn
colnames(d.p1) <- cn

# convert to long frames

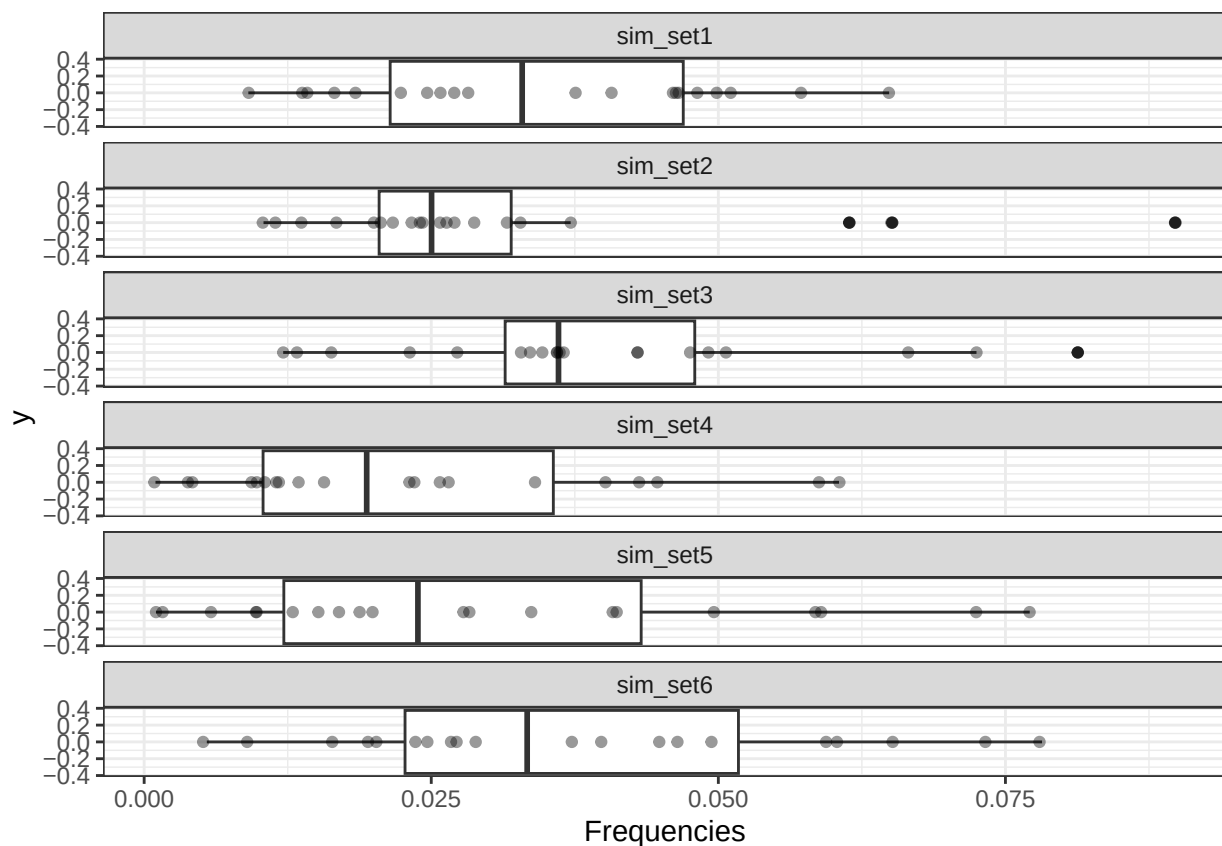
d.f1.long <- pivot_longer(d.f1,
  cols = all_of(cn),
  names_to = "sim_set",
  values_to = "Frequencies")

d.p1.long <- pivot_longer(d.p1,
  cols = all_of(cn),
  names_to = "sim_set",
  values_to = "Positives")

# create boxplots

freq.box <- ggplot(d.f1.long, aes(y= 0, x= Frequencies)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ sim_set, ncol = 1) +
  theme_bw();freq.box

```



```

ggsave("start_control_box_freq.png", plot = freq.box, dpi = 300, scale = 2)

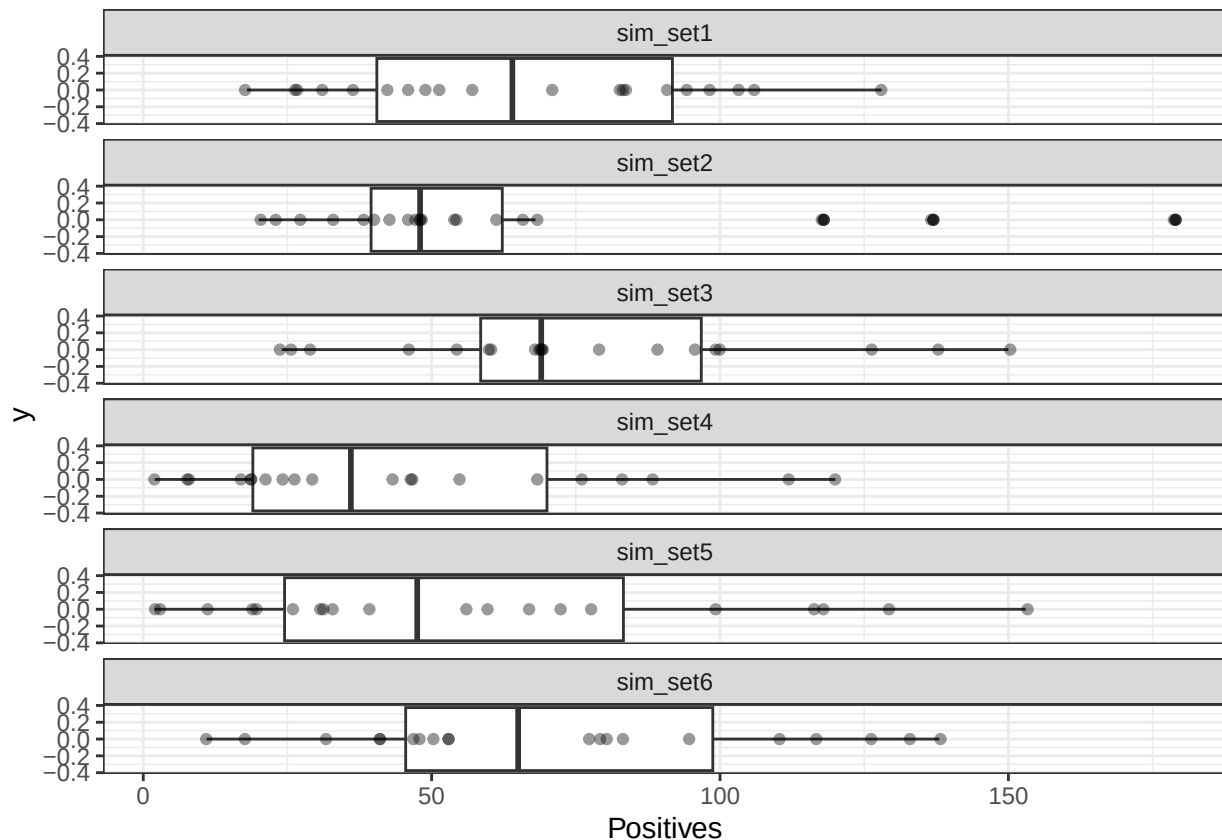
```

```

## Saving 13 x 9 in image

```

```
pos.box <- ggplot(d.p1.long, aes(y= 0, x= Positives)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ sim_set, ncol = 1) +
  theme_bw();pos.box
```



```
ggsave("start_control_box_pos.png", plot = pos.box, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

Centered Control

Next the kernel maximum and minimum x and y values were changed so that the simulations would be centered over the simulation window to ensure no simulations were being created where the cell assignments were being generated outside of the window. 20 simulations were run for 6 groups of simulations.

```
set.seed(800)
```

```
objs <- readRDS('StartControlCenteredOBS.rds')
```

Below are the parameters that were used for the first simulation. All of these parameters were kept constant.

```
param <- ExtractParameters(objs[[1]][[1]], 'All')
```

```
## Holes has not yet been simulated
```

```
pv <- vector('list', length(param))
for (i in 1:length(param)){
  pv[[i]] <- unlist(param[[i]])
}
```

```
names(pv) <- names(param)
pv
```

```
## $Tissue
##      k  xmin  xmax  ymin  ymax sdmin sdmax
## 10.0   0.0  10.0   0.0  10.0   0.5   2.0
##
## $Holes
##      xmin      xmax      ymin      ymax      sdmin      sdmax hole_prob1
##      0.00      10.00      0.00      10.00      0.50      2.00      0.20
## hole_prob2
##      0.35
##
## $Cells
##      Cell Type 1.k      Cell Type 1.xmin      Cell Type 1.xmax      Cell Type 1.ymin
##              1.0              2.5              7.5              2.5
##      Cell Type 1.ymax      Cell Type 1.sdmin      Cell Type 1.sdmax      Cell Type 1.probs1
##              7.5              0.5              2.0              0.0
## Cell Type 1.probs2
##              1.0
```

Calculations of frequencies/positivities and boxplots.

```
ca <- vector('list', length(objs))
cpos <- vector('list', length(objs))
cfreq <- vector('list', length(objs))

for (i in 1:length(objs)) {
  ca[[i]] <- vector('list', length(objs[[i]]))
  cfreq[[i]] <- vector('list', length(objs[[i]]))
  cpos[[i]] <- vector('list', length(objs[[i]]))
}

for (i in 1:length(objs)) {
  for(j in 1:length(objs[[i]])) {
    ca[[i]][[j]] <- CreateSpatialList(objs[[i]][[j]])[[1]][3:4]
    cfreq[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2')/sum(ca[[i]][[j]][1] == 'Tissue 2')
    cpos[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2')
  }
}

f1 <- vector('list', length(objs))
p1 <- vector('list', length(objs))
v.f1 <- vector('list', length(objs))
v.p1 <- vector('list', length(objs))
for (i in 1:length(f1)) {
  f1[[i]] <- unlist(cfreq[[i]])
  p1[[i]] <- unlist(cpos[[i]])
  v.f1[[i]] <- var(f1[[i]])
  v.p1[[i]] <- var(p1[[i]])
}

# create dataframes for pos and freq
```

```

d.f1 <- as.data.frame(f1)
d.p1 <- as.data.frame(p1)

cn <- list(paste0('sim_set', 1:6))[[1]]
colnames(d.f1) <- cn
colnames(d.p1) <- cn

# convert to long frames

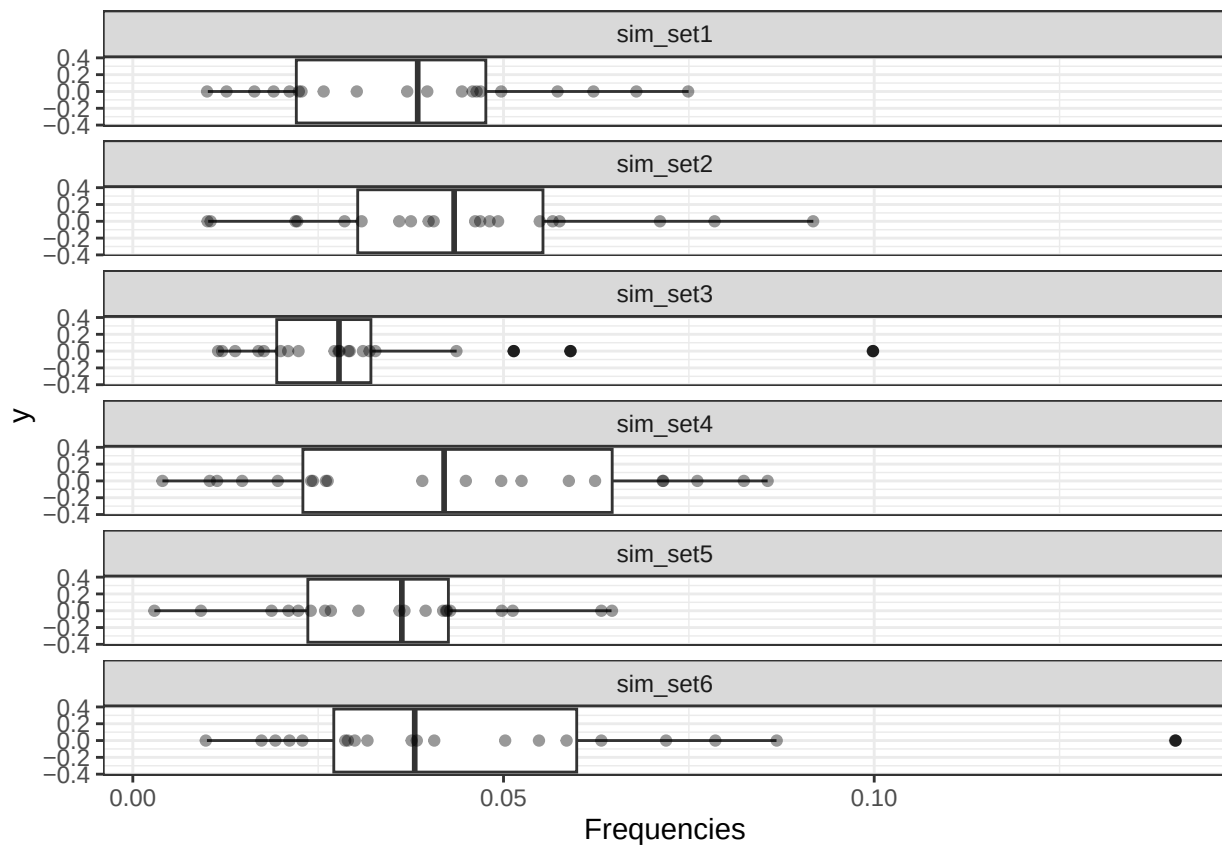
d.f1.long <- pivot_longer(d.f1,
                          cols = all_of(cn),
                          names_to = "sim_set",
                          values_to = "Frequencies")

d.p1.long <- pivot_longer(d.p1,
                          cols = all_of(cn),
                          names_to = "sim_set",
                          values_to = "Positives")

# create boxplots

freq.box <- ggplot(d.f1.long, aes(y= 0, x= Frequencies)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ sim_set, ncol = 1) +
  theme_bw();freq.box

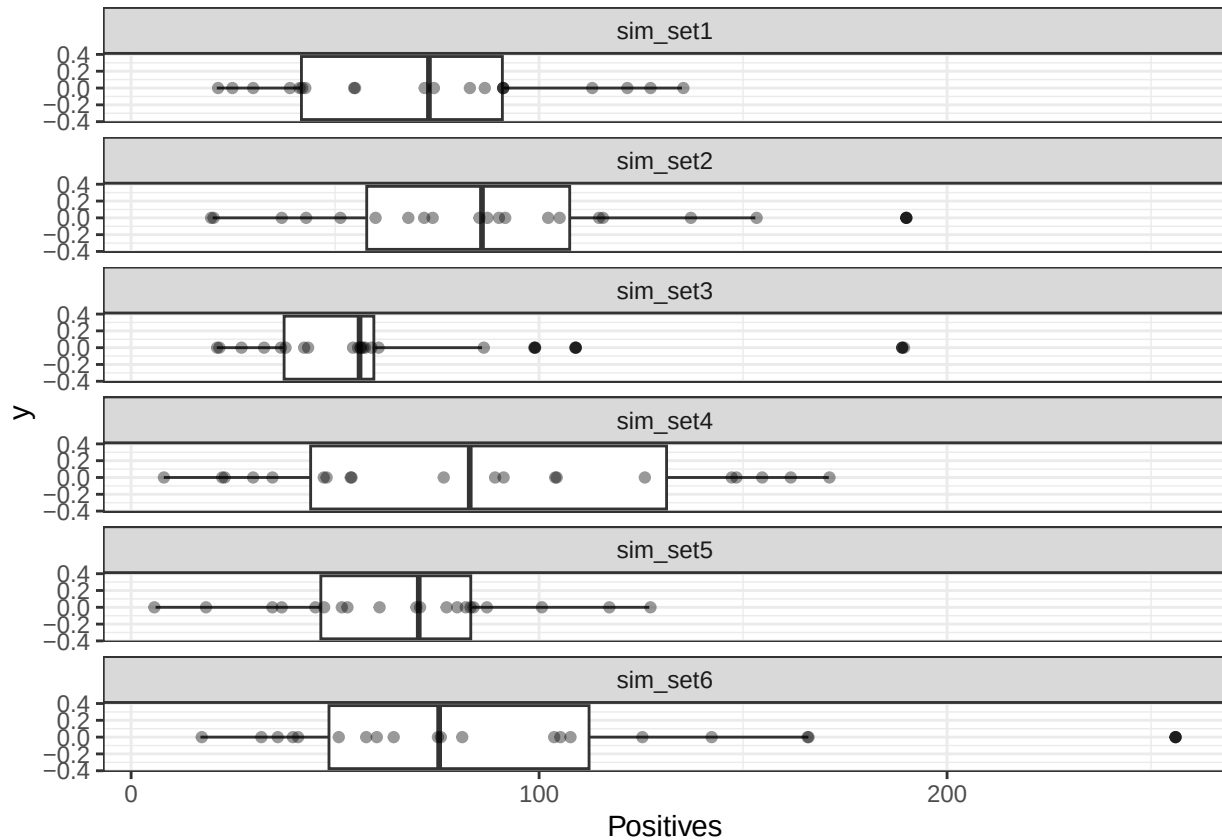
```



```
ggsave("start_centered_box_freq.png", plot = freq.box, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

```
pos.box <- ggplot(d.p1.long, aes(y= 0, x= Positives)) +  
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +  
  facet_wrap(~ sim_set, ncol = 1) +  
  theme_bw();pos.box
```



```
ggsave("start_centered_box_pos.png", plot = pos.box, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

Changing Starting Seeds

Next the seed was changed before each sset of simulations was created. 6 sets were created with 20 simulations created for each set. The same seed was used for the creation of each simulated object and the seed was changed before calling the generate cell positivity command.

```
s <- c(100, 200, 300, 400, 500, 600)
```

```
objs <- readRDS('StartSeeds10BJS.rds')
```

Below are the parameters that were used for the first simulation. All of these parameters were kept constant.

```
param <- ExtractParameters(objs[[1]][[1]], 'All')
```

```
## Holes has not yet been simulated
```

```

pv <- vector('list', length(param))
for (i in 1:length(param)){
  pv[[i]] <- unlist(param[[i]])
}
names(pv) <- names(param)
pv

```

```

## $Tissue
##      k  xmin  xmax  ymin  ymax sdmin sdmax
## 10.0   0.0  10.0   0.0  10.0   0.5   2.0
##
## $Holes
##      xmin      xmax      ymin      ymax      sdmin      sdmax hole_prob1
##      0.00      10.00      0.00      10.00      0.50      2.00      0.20
## hole_prob2
##      0.35
##
## $Cells
##      Cell Type 1.k  Cell Type 1.xmin  Cell Type 1.xmax  Cell Type 1.ymin
##                  1.0                0.0                10.0                0.0
##      Cell Type 1.ymax  Cell Type 1.sdmin  Cell Type 1.sdmax  Cell Type 1.probs1
##                  10.0                0.5                2.0                0.0
## Cell Type 1.probs2
##                  1.0

```

Calculations of frequencies/positivities and boxplots.

```

ca <- vector('list', length(objs))
cpos <- vector('list', length(objs))
cfreq <- vector('list', length(objs))

for (i in 1:length(objs)) {
  ca[[i]] <- vector('list', length(objs[[i]]))
  cfreq[[i]] <- vector('list', length(objs[[i]]))
  cpos[[i]] <- vector('list', length(objs[[i]]))
}

for (i in 1:length(objs)) {
  for(j in 1:length(objs[[i]])) {
    ca[[i]][[j]] <- CreateSpatialList(objs[[i]][[j]])[[1]][3:4]
    cfreq[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2')/sum(ca[[i]][[j]][1] == 'Tissue 2')
    cpos[[i]][[j]] <- sum(ca[[i]][[j]][2] == 1 & ca[[i]][[j]][1] == 'Tissue 2')
  }
}

f1 <- vector('list', length(objs))
p1 <- vector('list', length(objs))
v.f1 <- vector('list', length(objs))
v.p1 <- vector('list', length(objs))
for (i in 1:length(f1)) {
  f1[[i]] <- unlist(cfreq[[i]])
  p1[[i]] <- unlist(cpos[[i]])
  v.f1[[i]] <- var(f1[[i]])
  v.p1[[i]] <- var(p1[[i]])
}

```

```

}

# create dataframes for pos and freq
d.f1 <- as.data.frame(f1)
d.p1 <- as.data.frame(p1)

cn <- list(paste0('Seed', s))[[1]]
colnames(d.f1) <- cn
colnames(d.p1) <- cn

# convert to long frames

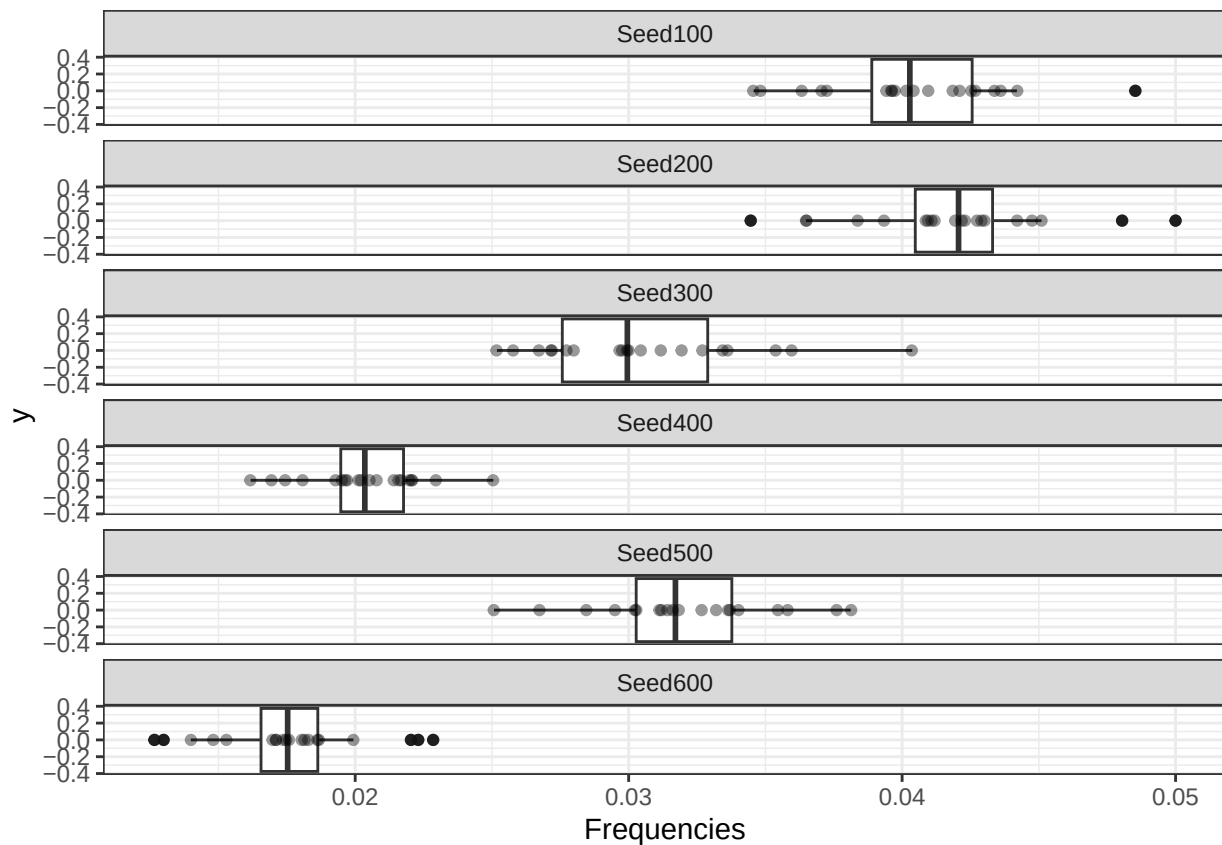
d.f1.long <- pivot_longer(d.f1,
                          cols = all_of(cn),
                          names_to = "Seed",
                          values_to = "Frequencies")

d.p1.long <- pivot_longer(d.p1,
                          cols = all_of(cn),
                          names_to = "Seed",
                          values_to = "Positives")

# create boxplots

freq.box <- ggplot(d.f1.long, aes(y= 0, x= Frequencies)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ Seed, ncol = 1) +
  theme_bw();freq.box

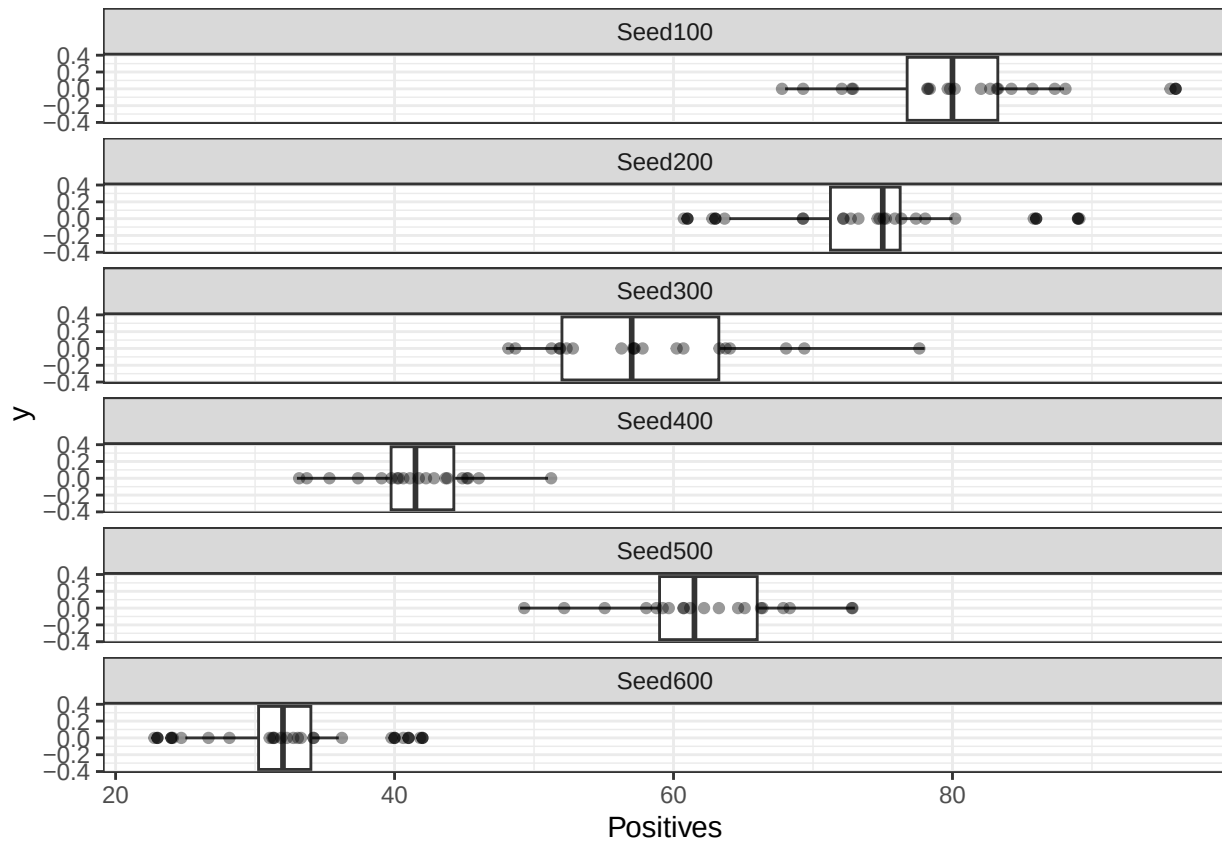
```



```
ggsave("start_seeds_box_freq.png", plot = freq.box, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

```
pos.box <- ggplot(d.p1.long, aes(y= 0, x= Positives)) +
  geom_boxplot() + geom_jitter(height= 0.000001, size= 1.5, color= 'black', alpha= 0.4) +
  facet_wrap(~ Seed, ncol = 1) +
  theme_bw();pos.box
```



```
ggsave("start_seeds_box_pos.png", plot = pos.box, dpi = 300, scale = 2)
```

```
## Saving 13 x 9 in image
```

THE FREQUENCIES SEEM TO BE MOST SENSITIVE TO CHANGES IN THE PROBABILITIES OF GENERATING POSITIVE CELL ASSIGNMENT TO CELL TYPE 1 (GPOSPROBS FOLDER)